KAIST SoC Summer Internship Research Plan

We request you to submit a research plan to encourage you to have a concrete research plan before you start your internship at KAIST. Discuss your research topic for your internship with your internship supervisor at KAIST before you start to fill out this form.

1. Student Information

Family Name	Suwananek	Given Name	Poopha
E-mail Address	auricridge@gmail.com	Mobile Phone	01067289535
Home University / Department		Chulalongkorn / Computer Engineering	
Internship Lab		Network Security and Privacy	
Internship Supervisor		Prof. Min Suk Kang	

In the following, provide your research plan including a title, objectives, a project summary, your role, and project schedule. Try to keep the content concise so that the document may <u>not exceed three pages</u>.

2. Title of the Project

The title should be concise yet informative, clearly representing the scope and objectives of the internship.

The incorporation of quantum key distribution within TLS

3. Objectives

Clearly define the goals and objectives of the internship. It's important to make sure these are S.M.A.R.T. (Specific, Measurable, Achievable, Relevant, and Time-Bound).

The goal of this project is to incorporate a quantum key distribution protocol for creating a symmetric session key within the TLS protocol. This incorporation and implementation could either completely replace or add to the already existing RSA algorithm (or other post quantum asymmetric encryption algorithm). The QKD protocol can be written using qiskit library and incorporate into OpenSSL's TLS. And then the new protocol can be benchmarked within simulated situations.

4. Project Summary

Provide a brief but detailed summary of the research project. Include information about the scope of the project, the problem it's addressing, and why it's important.

This project aims to address the potential security problem within TLS protocol (The security protocol which is used by HTTPS), specifically the creation and distribution of a symmetric session key. In TLS, session key is generate to be used after a client and a server had establish a valid connection, with the goal of encrypting any other data that will be communicate between the two. But in order to have a session key which is used by and is the same for both parties, they required a way to generate the key without leaking it. TLS does this by employing an asymmetric encryption (Public-Private key) such as RSA, etc. Unfortunately, with the recent development in quantum computing and optimization algorithms (Shor's algorithm), these asymmetric encryption could theoretically be broken. However, QKD protocol can be an alternative/addition to the session key generation dilemma. As QKD is theoretically secure against any computationally powerful adversary due to its inherent nature which utilizes quantum properties (No cloning theorem and bell's inequality), making it resilient to any attempts at eavesdropping or copying the key in an unsecure network and be immune to brute force tactics as the key is not secured by the virtue of being computationally hard to crack. But QKD itself can only securely create a session key but it alone cannot help prevents other kinds of vulnerabilities such as a-man-in-the-middle attack, etc. Therefore, the authentication and other security offered by TLS protocol is essential to cover where QKD could not. For this is the reason to incorporate QKD into TLS protocol. Other potential security measures that will be implemented with QKD and TLS are Device-Independent QKD for a no-trust environment and OpenSSL's API for the compatibility with current TLS implementations.

5. Roles and Responsibilities:

Define the specific roles and responsibilities that the intern will be expected to fulfill during the course of the internship.

To create a QKD protocol using qiskit.
To study OpenSSL TLS implementations.
To create DI-QKD protocol.
To incorporate QKD protocol for session key generation by either replacing or adding onto the already existing asymmetric encryption algorithm.
To explore Post Quantum encryptions and supplement QKD in session key generation.
To find vulnerability within the implementation.
To benchmark the new protocol in a simulated situation.

6. Schedule

Week	Task/Goal
1	To Prepare for the project. To familiarize to KAIST. To study OpenSSL TLS implementations.
2	To study OpenSSL TLS implementations. To create a QKD protocol using qiskit.
3	To create a QKD protocol using qiskit. To create DI-QKD protocol.
4	To create DI-QKD protocol. To incorporate QKD protocol for session key generation by either replacing or adding onto the already existing asymmetric encryption algorithm.

5	To explore Post Quantum encryptions and supplement QKD in session key generation. To find vulnerability within the implementation.
6	To find vulnerability within the implementation. To fix the vulnerability found within the implementation. To benchmark the new protocol in a simulated situation.
7	To find vulnerability within the implementation. To fix the vulnerability found within the implementation. To benchmark the new protocol in a simulated situation.
8	To find vulnerability within the implementation. To fix the vulnerability found within the implementation. To benchmark the new protocol in a simulated situation. To finish the project paper/report.

7. Signatures

Internship Student		Date	2025/06/25
Internship Supervisor		Date	2025/06/25